Daniel Gordon

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EDUCATION

Honours Computer Science and Management (B.Sci), Wilfrid Laurier University — September 2022 - Dec 2026
Research Interests: Deep Learning, Reinforcement Learning, Time-Series Forecasting, Optimization, and NLP — *Waterloo, ON*

MACHINE LEARNING

Research Assistant – AI/ML Research Project — September 2025 - Present
Wilfrid Laurier University — *Waterloo, ON*

- Reproduced and debugged a mathematical construction from a Russian research paper on vector-sequence transformations, identifying a flaw in the original Maple implementation that prevented full zero-vector convergence.
- **Applied a hybrid ML-driven error-localization approach**: used a lightweight supervised classifier trained on synthetic Maple traces (feature-engineered from symbolic expression depth, mutation count, and recurrence-pattern violations) to automatically flag suspicious recurrence steps. The model consistently highlighted the misaligned index-mapping region responsible for the error.
- Corrected the recurrence logic and implemented a validated Maple pipeline with repeated symbolic and numerical stress-testing (1000+ randomized trials) to confirm convergence. Compiled into a formal research report comparing theoretical behaviour with empirical outcomes.
- Reviewed related literature on symbolic/ML hybrid debugging to inform the classifier-based error-localization design.

N-Queens Reinforcement Learning Solver (Python, PyTorch) — January 2025 - April 2025
Wilfrid Laurier University — *Waterloo, ON*

- Built Q-learning and policy-gradient agents to converge on N-Queens solutions **up to 35× faster** than brute-force backtracking on $N \leq 20$.
- Implemented reward shaping, action masking, curriculum scaling; ran 40+ controlled experiments and validated stability with learning-curve statistical smoothing.
- Logged results using TensorBoard and performed ablation studies comparing Q-tables, linear function approximators, and shallow neural-policy networks.

EXPERIENCE

Contract Full Stack Developer — April 2024 - Present
Self-Employed — *Vaughan, ON*

- Built an AI-driven WhatsApp summarization platform using the WhatsApp Web API + **Gemini LLM**, serving ~200 daily users and processing >1M tokens/day of unstructured text.
- Designed end-to-end NLP pipelines for ingestion, batching, cleanup, summarization, and hallucination mitigation through iterative prompting.
- Optimized latency and context-window usage via dynamic chunking + caching strategies inspired by long-context LLM techniques, reducing average response time by ~30%.
- Developed modular REST APIs and React/Next.js interfaces used by 5+ clients. Delivered complete project lifecycles with consistent delivery.
- Implemented monitoring and logging tooling to analyze summarization failures, user behaviour, and model efficiency, informing iterative improvements to the NLP pipeline.

Full Stack / AI Developer — July 2025 - Present
MCG - Digital Assetization Company — *Vaughan, ON*

- Refactored a Python/JavaScript lip-sync ML pipeline for 3D avatars in Unity, improving phoneme-to-blendshape alignment and enabling both audio- and text-driven animation triggers.
- Prototyped end-to-end AI workflows: build Whisper-based speech-to-text and LangChain orchestration, benchmark alternatives, and recommend the best accuracy/latency configuration.
- Implemented model-level optimizations (quantization, beam-search tuning, batching), reducing inference latency ~25% without degrading accuracy.
- Expose lip-sync and AI capabilities as RESTful services and integrate with a React/Next.js front end to drive real-time avatar animations in the browser.

Junior Solution Architect and Developer — June 2025 - August 2025
LaunchPath Inc. — *Vaughan, ON*

- Architected a scalable system for a biosensor analytics platform, producing C4 diagrams, technical specifications, and end-to-end integration plans.
- Led Agile delivery across 3 sprints, defining epics/user stories and presenting sprint demos to stakeholders.
- Built a responsive React Native interface with secure integration to Python microservices for real-time biosensor data ingestion.

PROJECTS

- **TourSage – AI Tour Guide App:** Integrated generative AI + RAG (semantic search, embeddings) for itinerary generation using Python backends and Node/TypeScript.
- **wluNest Housing App:** Implemented search + geospatial queries.
- **Battleships AI Game (Java):** Implemented heuristic search with minimax-style decision trees and pruning heuristics.

SKILLS

Programming Languages:	Python, Java, JavaScript, C/C++, Lua, TypeScript, Dart, HTML, CSS, SQL, R, Bash
Technologies & Tools:	Git, Microsoft Office, React, React Native, Next.js, Flutter, VS, Apache Maven, Prisma, Tailwind, WireShark, MySQL, MongoDB, PostgreSQL, Figma, Framer, Node.js, Electron, Babylon.js, Flask, Unix, Expo, Unity, Hugging Face Transformers
Machine Learning:	PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, statsmodels, supervised learning, RL, sequence modeling, error analysis, benchmarking, ablations
Mathematical Foundation:	Linear Algebra (eigenvalues, SVD, PCA), Probability (Bayes, MLE), Optimization (gradient descent, convexity), Information Theory (entropy, KL divergence)